



OBJECT ORIENTED SOFTWARE ENGINEERING

Ms. Archana A

Department of Computer Applications

OBJECT ORIENTED SOFTWARE ENGINEERING

5. Unified Process Model

Archana A

Department of Computer Applications

Background:

- 1990--Many **object-oriented analysis** and **design** methods were **proposed**.
- They eventually worked together on a unified method, called the Unified Modeling Language (UML)
 - UML is a **robust notation** for the **modeling and development of object-oriented systems**.
 - UML became an industry standard in 1997.
 - However, UML does not provide the process framework, only the necessary technology for object-oriented development.

- Booch, Jacobson, and Rumbaugh later developed the **unified process**, which is a **framework** for object-oriented software engineering using UML
 - Draws on the **best features and characteristics** of conventional software process models.
 - Emphasizes the important role of software **architecture**.
 - Consists of a process flow that is **iterative and incremental**, thereby providing an **evolutionary feel**

- **Unified Process** — a “**use-case driven, architecture-centric, iterative and incremental**” software process closely aligned with the Unified Modeling Language (UML).
- It implements many of the **best principles of agile software development**.

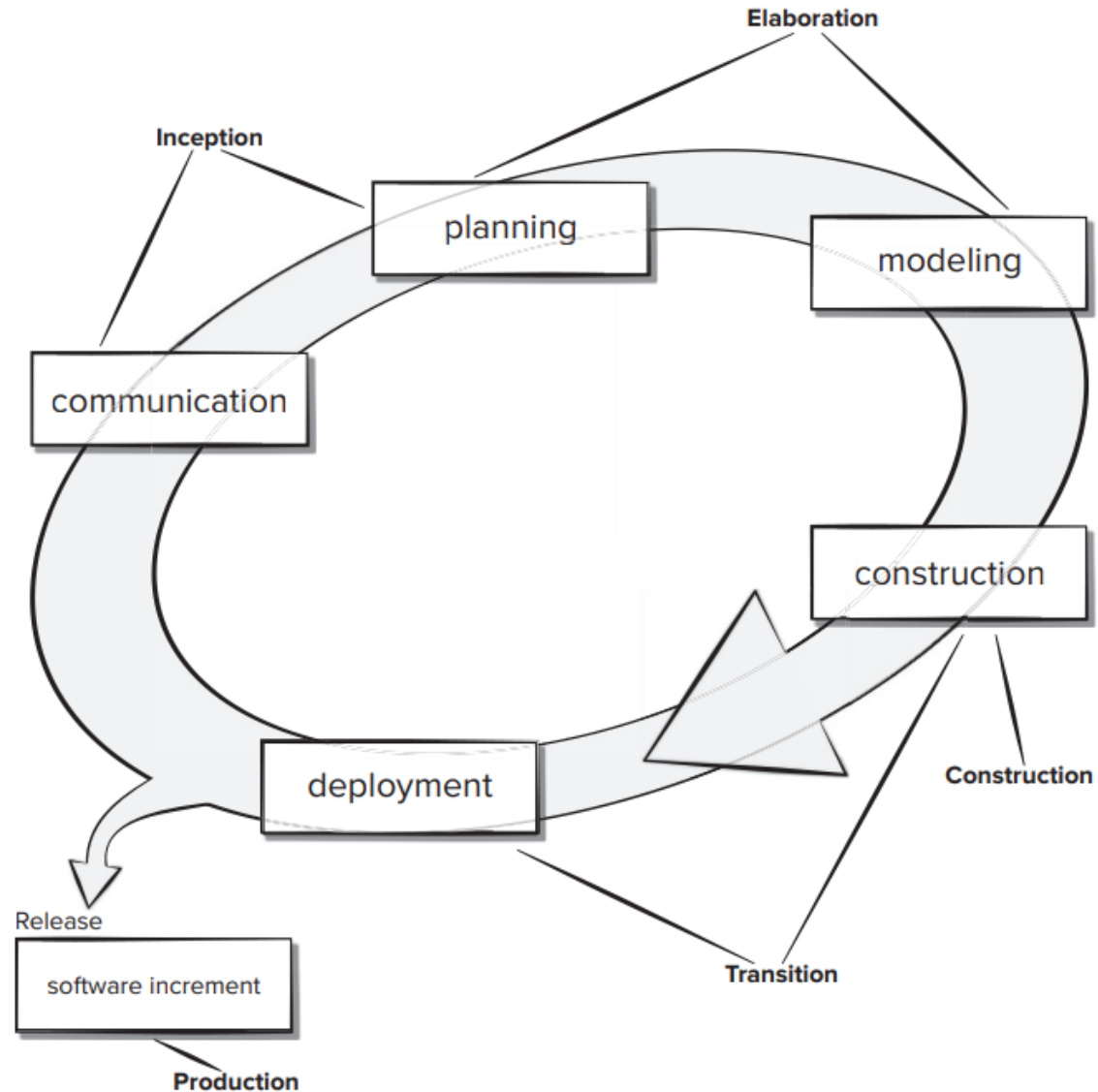
- The **Rational Unified Process®** is a *Software Engineering Process*. It provides a **disciplined approach** in assigning tasks and responsibilities in development process within an organization.
- Its goal is to ensure the **production of high-quality software** that meets the needs of its end-users, within a predictable **schedule and budget**.

The phases of Unified Process are:

1. The *Inception* phase
2. The *Elaboration* phase
3. The *Construction* phase
4. The *Transition* phase
5. The *Production* phase

OBJECT ORIENTED SOFTWARE ENGINEERING

The Phases of Unified Process Model...



Advantages:

- Well-documented and complete methodology.
- Open and Public.
- Training readily available.
- Changing Requirements.
- Reduced Integration Time and Effort.
- Higher Level of Re-use

Disadvantages:

- Process is too complex.
- Does not capture the sociological aspects of software development.



THANK YOU

Ms. Archana A

Department of Computer Applications

archana@pes.edu

+91 80 6666 3333 Extn 392